Synthetic example: constant speed

Combined question sheet and scoring rubric. Five criteria, six points. This is an invented training fixture, not a supplied instructor exam.

1(a). Which unit measures speed? Choose one: (1) m; (2) m/s; (3) s; (4) m/s^2.

2(a). A cart moves at a constant speed of 4 m/s for 3 s. Find the distance traveled, showing your relation.

2(b). At the same speed, how far does it travel in 9 s? Explain the time scaling; you may use your result from 2(a).

1.a.a - Speed unit - 1 point:

Award 1 only for an unambiguous final selection of option 2, m/s. Wrong, blank or multiple clearly uncancelled choices: 0. For a genuinely ambiguous cancellation, assign provisional 0 and flag review.

2.a.a - Distance relation - 1 point:

Award 1 for distance = speed times time, or an equivalent correct relation. Award 0.5 for an identifiable partly correct relation with a material omission; otherwise 0. A compact evaluated calculation can show the relation.

2.a.b - Distance evaluation - 1 point:

Award 1 for 12 m with correct arithmetic and units. Partial credit for correct identifiable content with a material omission is allowed within 0..1. This does not remove the independent relation point.

2.b.a - Time scaling - 1 point:

Award 1 for identifying that the time (and distance at constant speed) triples; an explicit multiplication by 3 suffices. Otherwise 0.

2.b.b - Later distance - 2 points:

Award up to 2 for the later distance: 36 m from correct data, or a correctly tripled earlier distance under follow-through. Use increments of 0.5 for identifiable partially correct content. Do not repeat a deduction for an earlier arithmetic error.